

Sustainable Development, Equity & the Triple Bottom Line

The three ideas underneath every sustainability framework you will ever use

**Every sustainability report, every ESG score, every
climate lawsuit
relies on three ideas most people can't actually define.**

*By the end of today, you will be able to define all three precisely — and use them to judge whether a real decision is
actually sustainable.*

Three Ideas, Built One on Top of the Other



01

Sustainable Development

The umbrella concept: meeting present needs without compromising the future.



02

Intergenerational Equity

The ethical engine inside that concept: fairness to generations not yet born.



03

The Triple Bottom Line

The measurement tool that turns the concept into something you can actually score.

SECTION 1 OF 3

Sustainable Development

The umbrella concept that gave the entire sustainability movement its shared vocabulary.



The Brundtland Definition

The single most quoted sentence in the history of the sustainability movement.

*"Development that meets the needs of the present
without compromising the ability of future generations to meet their own
needs."*



The Source

"Our Common Future" (1987), also called the Brundtland Report — published by the UN World Commission on Environment and Development (WCED).



The Chair

Chaired by Gro Harlem Brundtland, former Prime Minister of Norway — the commission was convened by the UN in 1983.



The Process

Nearly 900 days of research, public hearings, and consultation across 21 commissioner countries before publication in April 1987.

The Two Ideas Hidden Inside the Definition

The Brundtland Report itself immediately unpacked its own famous sentence into two essential concepts — both are usually forgotten.

The Concept of "Needs"

In particular, the essential needs of the world's poor, to which overriding priority should be given. Sustainable development is not neutral between rich and poor — it explicitly prioritizes basic human needs first.

The Idea of "Limitations"

Limitations imposed by the state of technology and social organization on the environment's ability to meet present and future needs. Sustainability is not a fixed ceiling — better technology and institutions can raise it.

The Three Pillars of Sustainable Development

Nearly every sustainability framework since 1987 — including the Triple Bottom Line we cover later today — traces back to this three-way balance.



Environmental

Protecting ecosystems, biodiversity, and natural resources so they remain available and functional over time.



Social

Ensuring equity, health, education, and quality of life across all people — not just economic output.



Economic

Generating prosperity and livelihoods in a way that can continue indefinitely, not just in the short term.

Core Principles of Sustainable Development

Beyond the definition, a small set of legal and policy principles operationalize sustainable development in practice.

The Precautionary Principle

Where there are threats of serious or irreversible environmental damage, lack of full scientific certainty should not be used as a reason to postpone preventive measures.

The Polluter Pays Principle

The party responsible for producing pollution should bear the cost of managing it, rather than passing that cost on to society or the environment.

Intergenerational Equity

Fairness between present and future generations — significant enough to be Section 2 of today's lecture in its own right.

Public Participation

Environmental decisions should involve the people affected by them, with access to information and justice.

Sustainable Development in Indian Law

Indian courts adopted these principles into binding law years before most other jurisdictions.

Vellore Citizens Welfare Forum v. Union of India (1996)

The Supreme Court of India formally held that the Precautionary Principle and the Polluter Pays Principle are part of Indian environmental law.

Article 21 of the Constitution

The Supreme Court has repeatedly interpreted the fundamental right to life to include the right to a clean and healthy environment.

The Public Trust Doctrine

Indian courts (e.g., M.C. Mehta v. Kamal Nath, 1997) have held that certain natural resources are held by the state in trust for public use, including for future generations.

According to the Brundtland Report itself, sustainable development gives "overriding priority" to which group's needs?

- A Shareholders of large multinational corporations
- B The essential needs of the world's poor** ✓
- C Future generations exclusively, over the present generation
- D National governments' economic growth targets

ANSWER & WHY

B. The essential needs of the world's poor

The Brundtland Report is explicit that the concept of "needs" centers "in particular the essential needs of the world's poor, to which overriding priority should be given" — sustainable development was never meant to be neutral between rich and poor.

SECTION 2 OF 3

Intergenerational Equity

The ethical engine hidden inside "sustainable development" — fairness to people who cannot yet speak for themselves.

What Is Intergenerational Equity?

A principle of fairness that extends across time, not just across a single population.

The Core Idea

Each generation holds the Earth in trust for those that follow — present generations should not use up or degrade resources in ways that leave future generations worse off.

Why "Equity," Not Just "Sustainability"

It reframes sustainability as a justice question: future people cannot vote, lobby, or buy carbon offsets today — someone has to represent their interests now, on their behalf.

Where It Comes From

Formalized as an international legal theory by Professor Edith Brown Weiss in her 1989 book, "In Fairness to Future Generations."

Edith Brown Weiss's Three Principles

A precise, widely cited framework for what one generation owes the next.



Conservation of Options

Each generation should conserve the diversity of the natural and cultural resource base, so it does not unduly restrict the options available to future generations.



Conservation of Quality

Each generation should maintain the quality of the planet so that it is passed on in no worse condition than it was received.



Conservation of Access

Each generation should provide its members equitable rights of access to the legacy of past generations, and conserve this access for future generations.

Intergenerational vs. Intragenerational Equity



Intragenerational Equity

Fairness among people alive today — rich nations vs. poor, one community vs. another.

Example: should India and the US bear equal responsibility for emissions cuts today?

Addressed by mechanisms like climate finance and "loss and damage" funds.

Both are required for sustainable development — one without the other is incomplete.



Intergenerational Equity

Fairness between generations across time — the present vs. those not yet born.

Example: should today's generation deplete oil reserves that future generations may need more?

Addressed by mechanisms like resource conservation, long-term planning, and discount-rate policy.

Both are required for sustainable development — one without the other is incomplete.

Why the Discount Rate Is an Ethical Choice

GIVEN

- A policy prevents \$1,000,000 of climate damage occurring in the year 2100 (100 years away)
- Stern Review (2006) used a low discount rate of approximately 1.4% per year
- A higher, more traditional market-based discount rate: approximately 5% per year

FORMULA

$$\text{Present Value} = \text{Future Value} \div (1 + r)^t$$

STEP-BY-STEP CALCULATION

- 1 At Stern's ~1.4% rate: $\$1,000,000 \div (1.014)^{100} \approx \$249,000$ in today's money
- 2 At a ~5% rate: $\$1,000,000 \div (1.05)^{100} \approx \$7,600$ in today's money
- 3 The identical future benefit is worth roughly 33× more under the low discount rate than the high one.

33× Difference

Same future benefit, same year, same policy — but the choice of discount rate alone changes whether climate action looks "worth it" today by more than an order of magnitude.

Intergenerational Equity in the Courts

What began as ethical theory is now active, binding litigation strategy worldwide.

M.C. Mehta v. Union of India (multiple cases)

Indian courts have repeatedly invoked intergenerational equity alongside the public trust doctrine to justify environmental protection orders.

Urgenda Foundation v. State of the Netherlands (2019)

The Dutch Supreme Court ordered the government to cut emissions faster, citing its duty of care to both current and future citizens.

Youth-Led Climate Litigation Worldwide

Cases brought by young people in multiple countries have argued that inadequate climate policy violates the constitutional rights of generations who will live with its consequences.

"Youth climate litigation asks courts to protect people who are not yet born, or too young to vote. Should judges have that power — or should it stay with elected governments?"

There is no clean answer here. Notice the tension between democracy and long time horizons.

📄 3 MIN

Which of Edith Brown Weiss's three principles is most directly violated by permanently destroying a unique ecosystem with no equivalent elsewhere?

A Conservation of Options



B Conservation of Quality

C Conservation of Access

D None of the three — biodiversity is outside this framework

ANSWER & WHY

A. Conservation of Options

Conservation of Options requires preserving the diversity of the natural resource base so future generations retain choices. Permanently destroying a unique, irreplaceable ecosystem directly forecloses options no future generation can ever recover.

SECTION 3 OF 3

The Triple Bottom Line

The measurement tool that turns "sustainable development" from a philosophy into a scorecard.



What Is the Triple Bottom Line?

A simple, radical idea: a company's success should never be measured on profit alone.

People · Planet · Profit



The Coinage

John Elkington, founder of the consultancy SustainAbility, coined the term "triple bottom line" in 1994.



The Book

Popularized in his 1997 book, "Cannibals with Forks: The Triple Bottom Line of 21st Century Business."



The Argument

Companies should report three separate bottom lines — economic, social, environmental — not just the financial one.

The Economic Bottom Line ("Profit")

Not just whether a company makes money — but whether it does so in a way that can continue indefinitely.

Traditional Financial Performance

Revenue, profit margins, return on investment — the metrics every business already tracks.

Long-Term Economic Viability

Can the business model survive resource price shocks, regulatory change, and reputational risk over decades, not just this quarter?

Value Created for the Wider Economy

Wages paid, taxes contributed, suppliers supported — economic value the company generates beyond its own balance sheet.

The Social Bottom Line ("People")

How an organization treats the humans inside it, and the communities around it.

Fair Labour Practices

Living wages, safe working conditions, freedom of association — and, across supply chains, freedom from forced or child labour.

Community Impact

Effects on local health, housing, and livelihoods — both positive (local employment) and negative (pollution, displacement).

Diversity, Equity & Human Rights

Fair treatment and opportunity regardless of gender, caste, ethnicity, or background, within the organization and its value chain.

The Environmental Bottom Line ("Planet")

The ecological footprint of an organization — everything covered in the rest of this course, in miniature.

Resource Use & Emissions

Energy and water consumption, waste generated, and greenhouse gas emissions across Scope 1, 2, and 3.

Pollution & Ecosystem Impact

Effects on air and water quality, biodiversity, and land use — both direct operations and the full supply chain.

Circularity & Resource Efficiency

How much of what the organization uses is renewable, recycled, or designed to be reused, rather than extracted and discarded.

How the Triple Bottom Line Gets Measured

The idea only matters if it can actually be scored — these are the leading frameworks that do it.



GRI Standards

The Global Reporting Initiative — the most widely used framework worldwide for sustainability disclosure across all three pillars.



ESG Scores

Environmental, Social, and Governance ratings used by investors to price climate and social risk into company valuations.



B Corp Certification

A rigorous third-party certification verifying a company meets high standards of social and environmental performance, alongside profit.



India's BRSR

SEBI's Business Responsibility and Sustainability Reporting mandate — India's own binding version of triple-bottom-line disclosure.

When a Company Restructured Around All Three Lines



Patagonia — "Earth Is Now Our Only Shareholder" (2022)

Founder Yvon Chouinard transferred 100% of Patagonia's voting stock to a specially designed trust, and all non-voting stock to a nonprofit — directing all profits not reinvested in the business toward fighting the climate crisis.

Economic ("Profit")

Patagonia remains a for-profit business, generating revenue to fund its own operations and growth — profit was never eliminated, only redirected.

Social ("People")

Long-standing fair labor and supply-chain standards, plus a governance structure explicitly designed to resist pressure to prioritize shareholder returns.

Environmental ("Planet")

An estimated ~\$100 million/year in profit now flows to environmental causes and climate action instead of private shareholders.

The Lesson

A real example of ownership structure itself — not just a sustainability report — being redesigned around all three bottom lines simultaneously.

Even Its Inventor Has Doubts

In 2018, John Elkington published a Harvard Business Review article publicly "recalling" his own 25-year-old concept.

"The Triple Bottom Line has failed to bury the single bottom line paradigm... It was never supposed to be just an accounting system."

The Original Ambition

Elkington intended TBL to drive systemic transformation of capitalism itself — not just a reporting checklist.

What Actually Happened

Many companies reduced it to an "accounting tool" — adding social and environmental metrics to reports without changing core business models.

The Quantification Problem

Environmental and social impact are genuinely hard to express in the same monetary units as profit, inviting selective or superficial reporting.

The Greenwashing Risk

A polished sustainability report can create an appearance of responsibility that isn't backed by real change — the exact critique this course applies to weak carbon credits.

Which Bottom Line Does This Improve?

For each company action, decide which of the three bottom lines it primarily strengthens.

		ECONOMIC	SOCIAL	ENVIRONMENTAL
1	Raising factory wages above the local living wage			
2	Switching a factory to renewable electricity	3	1	2
3	Diversifying suppliers to reduce financial risk	6	4	5
4	Publishing a supply-chain human rights audit			
5	Redesigning packaging to cut material waste			
6	Cutting costs through improved operational efficiency			

In his 2018 Harvard Business Review article, what did John Elkington argue had happened to the Triple Bottom Line?

- A It had been banned by most stock exchanges
- B It had been reduced to an accounting tool instead of driving systemic change ✓**
- C It had been fully achieved by all Fortune 500 companies
- D It had been replaced entirely by the SDGs

ANSWER & WHY

B. It had been reduced to an accounting tool instead of driving systemic change

Elkington argued that TBL was intended as a "genetic code" to transform capitalism itself, but in practice was reduced by many companies to a reporting and accounting exercise — prompting him to publicly "recall" the concept for a rethink.

BRINGING IT TOGETHER

How the Three Ideas Actually Connect

Each concept plays a distinct, necessary role — none of the three is sufficient on its own.

Sustainable Development

The umbrella goal

Sets the overall destination: meeting present needs without compromising the future.

Intergenerational Equity

The ethical justification

Explains why the future matters morally, and gives it a voice in present-day decisions.

The Triple Bottom Line

The measurement tool

Turns the goal into something scoreable — three concrete lines a real organization can report against.

"A mining company proposes a new project: strong local jobs and tax revenue, but it will deplete a resource future generations cannot replace. Using all three frameworks from today, how would you evaluate it?"

Consider: Brundtland's "needs," Weiss's conservation principles, and all three TBL pillars.

🕒 4 MIN

Three Ideas, Recapped

1

The Brundtland Definition

Development meeting present needs without compromising the ability of future generations to meet their own.

2

Two Hidden Concepts

"Needs" (prioritizing the world's poor) and "limitations" (imposed by technology and social organization).

3

Three Pillars of SD

Environmental, Social, and Economic — the same three dimensions that reappear in the TBL.

4

SD in Indian Law

The Precautionary Principle and Polluter Pays Principle are binding law since Vellore Citizens (1996).

5

Intergenerational Equity

Fairness across time, not just across today's population — formalized by Edith Brown Weiss (1989).

6

Weiss's Three Principles

Conservation of Options, Quality, and Access — what one generation owes the next.

7

The Discount Rate Debate

A single technical choice can change the value of future harm by 30× or more — a hidden ethical decision.

8

The Triple Bottom Line

People, Planet, Profit — coined by Elkington (1994), popularized in "Cannibals with Forks" (1997).

9

The Elkington "Recall"

Even TBL's inventor warned it had become an accounting checklist instead of driving real transformation.

10

The Synthesis

Sustainable development is the goal, intergenerational equity is the ethics, and the Triple Bottom Line is the scorecard.

Principles of Sustainability

Ecological Integrity · Resource Efficiency · Social Equity · Economic Viability

**"Sustainable" is not one property.
It is four different tests a decision has to pass, all at
once.**

Today gives you the actual engineering and scientific vocabulary behind each test — the tools you'll use to judge whether something is genuinely sustainable, or just sounds like it.

Four Principles, One Sustainability Test

These four principles sharpen the three Triple Bottom Line pillars from our last lecture into precise, usable engineering and policy criteria.



01

Ecological Integrity

Are ecosystems kept functioning, resilient, and intact?



02

Resource Efficiency

Is output achieved with less material, energy, and waste?



03

Social Equity

Are benefits and burdens shared fairly across all people?



04

Economic Viability

Can this actually continue, at its true, full cost?

SECTION 1 OF 4

Ecological Integrity

Whether the natural systems we depend on can keep functioning — not just whether individual species survive.



What Is Ecological Integrity?

More than "protecting nature" — a precise, measurable property of an ecosystem.

The Working Definition

The maintenance of an ecosystem's composition (what species are present), structure (how they're organized), and function (nutrient cycling, energy flow) across time.

Why "Integrity," Not Just "Health"

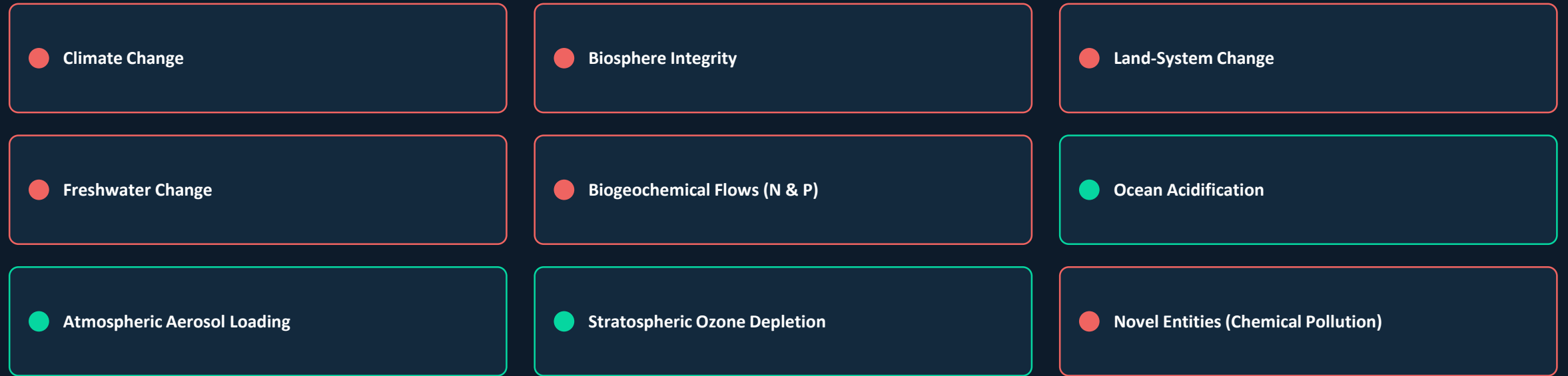
Integrity specifically includes resilience — an ecosystem's ability to absorb disturbance and recover, not just its condition at a single snapshot in time.

The Practical Test

A system has ecological integrity if it can keep delivering its ecosystem services — clean water, fertile soil, climate regulation — under realistic future stress.

The Planetary Boundaries Framework

Nine processes that regulate the stability of the entire Earth system — proposed by Johan Rockström and colleagues at the Stockholm Resilience Centre.



● Transgressed (6 of 9) ● Within safe limits (3 of 9)

As of the 2023 Richardson et al. assessment (Science Advances), six of nine boundaries are transgressed — later tracking suggests a seventh (ocean acidification) may also now be crossed.

Source: Rockström et al. 2009; Steffen et al. 2015; Richardson et al. 2023, Science Advances.

Ecosystem Services We Depend On

Ecological integrity is not an abstract environmental concern — it is the infrastructure the entire economy runs on, for free.



Pollination

Roughly three-quarters of global food crop types depend at least partly on animal pollinators — a free service with no engineered substitute at scale.



Water Filtration

Wetlands and forested watersheds naturally filter pollutants and regulate water flow, avoiding costly built infrastructure.



Carbon Sequestration

Forests, soils, and oceans absorb a substantial share of human CO₂ emissions every year, at zero cost — until they're degraded.



Soil Fertility

Healthy soil ecosystems cycle nutrients that make agriculture possible at all — a service that took millennia to build and can be lost in years.

How Ecological Integrity Gets Measured

Turning "is nature healthy?" into a number.



Species Richness & Abundance

How many species are present, and in what numbers — a direct signal of ecosystem diversity and stability.



Habitat Connectivity

Whether habitats are fragmented into isolated patches, or connected enough for species to migrate, breed, and adapt.



Resilience & Recovery Time

How quickly a system returns to its prior state after a disturbance like fire, drought, or flood — the core of "integrity" itself.

"Of the six already-transgressed planetary boundaries, which do you think gets the least public attention relative to how serious it is — and why?"

Climate change dominates headlines. What about the other five?

🕒 3 MIN

According to the 2023 Richardson et al. assessment, how many of the nine planetary boundaries have been transgressed?

A Three of nine

B Four of nine

C Six of nine



D All nine

ANSWER & WHY

C. Six of nine

The landmark 2023 study (Richardson et al., Science Advances) was the first to quantify all nine boundaries, finding six already transgressed: climate change, biosphere integrity, land-system change, freshwater change, biogeochemical flows, and novel entities.

SECTION 2 OF 4

Resource Efficiency

Achieving the same, or better, outcomes while extracting, consuming, and wasting less.



What Is Resource Efficiency?

Getting more human wellbeing and economic output out of every unit of material and energy consumed.

The Working Definition

Maximizing the value delivered — goods, services, wellbeing — per unit of natural resource extracted, used, and discarded.

The Central Goal: Decoupling

Breaking the historical link between economic growth and resource use or environmental harm, so the economy can grow while its footprint shrinks.

Relative vs. Absolute Decoupling

Relative decoupling: resource use grows slower than the economy. Absolute decoupling: resource use falls even as the economy grows — the much harder, and more necessary, goal.

The IPAT Equation: Reasoning About Impact

GIVEN

- Population (P) grows by 10% over a decade
- Affluence (A) — GDP per capita — grows by 20% over the same decade
- Technology (T) — environmental impact per unit of GDP — improves (falls) by 25%

FORMULA

$$\text{Impact} = \text{Population} \times \text{Affluence} \times \text{Technology}$$
$$(I = P \times A \times T)$$

STEP-BY-STEP CALCULATION

- 1 Express each factor as a multiplier: $P = 1.10$, $A = 1.20$, $T = 0.75$ (a 25% cut in impact intensity)
- 2 Multiply: $I = 1.10 \times 1.20 \times 0.75 = 0.99$
- 3 Net result: total environmental impact stays roughly flat — even slightly down — despite real growth in both population and wealth.

$$I \approx 0.99 \times$$

Technology improvement alone offset a decade of population and income growth — this is exactly what "decoupling" looks like in the equation that first formalized the idea (Ehrlich & Holdren, 1970s).

Linear vs. Circular Economy



Linear Economy ("Take-Make-Dispose")

Resources are extracted, used once, and discarded as waste at end of life.

Value is created once, then lost — the default model of most 20th-century industry.

"Waste" is treated as an unavoidable, external cost.

Defined by contrast in the Ellen MacArthur Foundation's widely cited circular economy framework.



Circular Economy

Products and materials are designed to be reused, repaired, remanufactured, or recycled.

Value is retained in the system for as long as possible — waste is designed out from the start.

One process's waste becomes another process's input — "industrial symbiosis."

Defined by the Ellen MacArthur Foundation as: design out waste, keep materials in use, regenerate natural systems.

Factor 4 and Factor 10: How Much Efficiency Is Enough?

Two influential targets for how far resource efficiency needs to go.

Factor 4

Proposed by von Weizsäcker, Lovins & Lovins (1997): doubling wealth while halving resource use — a 4× improvement in resource productivity.

Factor 10

A more radical target proposed by Friedrich Schmidt-Bleek: wealthy nations cutting resource use by 90% (10× more efficient), to leave ecological room for developing nations to grow.

Why Set Numerical Targets At All

Vague commitments to "be more efficient" are unfalsifiable. A stated factor gives engineers and policymakers something concrete to design against.

The Jevons Paradox

Efficiency does not automatically reduce total consumption — sometimes it does the opposite.

In 1865, economist William Stanley Jevons observed that more efficient steam engines did not reduce Britain's coal consumption — they increased it, by making coal power cheaper and more widely used.

The Mechanism

Efficiency lowers the effective cost of using a resource — and lower cost usually increases demand, sometimes enough to outweigh the efficiency gain entirely.

A Modern Example

More fuel-efficient cars can lead people to drive more miles, or buy larger vehicles, partially or fully offsetting the fuel savings ("the rebound effect").

The Lesson For This Course

Resource efficiency (Factor 4, Factor 10, IPAT's "T" term) is necessary — but without a hard cap or price signal alongside it, efficiency gains can be consumed away.

Circular or Linear Economy Practice?

For each business practice, decide whether it reflects circular or linear economic thinking.

- 1 Designing products for easy disassembly and repair
- 2 Single-use plastic packaging sent to landfill
- 3 One factory's waste heat powering a neighboring factory
- 4 Planned obsolescence to drive repeat purchases
- 5 Remanufacturing used industrial equipment for resale
- 6 Landfilling end-of-life electronics

CIRCULAR

LINEAR

1

2

3

4

5

6

What does "Factor 4" refer to in the resource efficiency literature?

- A Reducing a country's population by a factor of four
- B Doubling wealth while halving resource use — a 4× improvement in resource productivity ✓**
- C A four-year plan to eliminate all industrial waste
- D Reducing GDP by a factor of four to protect the environment

ANSWER & WHY

B. Doubling wealth while halving resource use — a 4× improvement in resource productivity

Factor 4, proposed by von Weizsäcker, Lovins & Lovins (1997), specifically means doubling wealth while halving resource use — a 4× gain in how much value is extracted per unit of resource, not a reduction in wealth itself.

SECTION 3 OF 4

Social Equity

Whether the benefits and burdens of development and environmental change are shared fairly.



What Is Social Equity in Sustainability?

Not simply "equality" — fairness that accounts for different starting points and different exposure to harm.

Distributive Justice

Who actually receives the benefits of development — jobs, infrastructure, clean energy — and who bears its costs, like pollution or displacement?

Procedural Justice

Who gets a genuine say in decisions that affect their environment and livelihood — and who is left out of the room entirely?

Why It Belongs Alongside Ecology and Economics

A technically efficient, profitable, low-emission project that devastates one community's livelihood is not sustainable — it has simply moved the cost onto people with the least power to resist it.

Environmental Justice: Where the Idea Comes From



Warren County, North Carolina (1982)

Protests against a PCB toxic waste landfill sited in a majority-Black, low-income county are widely credited as the birth of the environmental justice movement, later formalized in the influential 1987 "Toxic Wastes and Race" report.



Bhopal Gas Tragedy, India (1984)

A gas leak at a Union Carbide pesticide plant killed thousands immediately and affected several hundred thousand residents of a densely populated, low-income neighborhood — still cited as one of the world's worst industrial disasters and a landmark environmental justice case.

The pattern in both cases: hazards are disproportionately sited near communities with the least political and economic power to resist them.

Equity in Climate Impacts

~50%

Share of global consumption-based emissions attributed to the wealthiest 10% of the world's population

~10%

Share of global consumption-based emissions attributed to the poorest half of the world's population

3.3–3.6 Bn

People already living in contexts highly vulnerable to climate impacts (IPCC AR6)

The people contributing least to the problem are consistently exposed to the most severe consequences of it — the central equity fact of the entire climate crisis.

Source: Oxfam Carbon Inequality reports (illustrative synthesis); IPCC AR6 (2022).

"Bhopal happened in 1984. Can you think of a more recent example — anywhere in the world — where an environmental hazard was disproportionately sited near a poor or marginalized community?"

This pattern did not end in the 1980s.

📺 3 MIN

The 1982 protests widely credited as the origin of the environmental justice movement took place against a PCB landfill in which US county?

A Los Angeles County, California

B Warren County, North Carolina



C Cook County, Illinois

D Harris County, Texas

ANSWER & WHY

B. Warren County, North Carolina

Protests against a PCB toxic waste landfill sited in majority-Black, low-income Warren County, North Carolina, in 1982 are widely credited as the founding moment of the US environmental justice movement.

SECTION 4 OF 4

Economic Viability

Whether an activity can genuinely continue — once its full, true cost is actually counted.



What Is Economic Viability in Sustainability?

Not the same question as "is it profitable right now?"

The Working Definition

Economic activity that can be sustained over the long term without depleting the natural and social capital it depends on to keep functioning.

Why Short-Term Profit Isn't Enough

A business can be highly profitable while quietly depleting the soil, water, or community trust it needs to operate in ten years — profitable today, insolvent against reality tomorrow.

The Core Requirement: Internalizing Externalities

Prices must reflect the true cost of an activity — including environmental and social costs — or the market will systematically favor the least sustainable option.

True Cost Accounting: Pricing an Externality

GIVEN

- A product's market price: ₹100 per unit
- Uncompensated pollution and health costs borne by the surrounding community: ₹30 per unit
- This gap between market price and full cost is called an externality

FORMULA

$$\text{True Cost} = \text{Market Price} + \text{Externalized Cost}$$

STEP-BY-STEP CALCULATION

- 1 Add the externality to the market price: $₹100 + ₹30 = ₹130$
- 2 The real cost to society is 30% higher than what the market price actually reflects
- 3 This is the same logic behind the Social Cost of Carbon (~\$190/tonne, US EPA, 2023) covered in our carbon pricing lecture.

₹130 True Cost

The community is effectively subsidizing the product by ₹30 per unit through unpriced pollution and health impacts — a hidden transfer the market price never shows.

Metrics That Go "Beyond GDP"

GDP counts all economic activity as positive — including activity that depletes the future. These frameworks try to correct for that.



Green GDP

Subtracts the estimated cost of environmental degradation and resource depletion from conventional GDP figures.



Genuine Progress Indicator (GPI)

Adjusts GDP for income distribution, environmental costs, and unpaid work like caregiving — capturing what GDP misses entirely.



Gross National Happiness (Bhutan)

Bhutan's own wellbeing-centered alternative to GDP, formally adopted as national policy, weighing culture, environment, and governance alongside economic output.



Human Development Index (HDI)

The UNDP's composite measure combining health, education, and income — used globally as a richer alternative to income alone.

Which country pioneered "Gross National Happiness" as a formal alternative to GDP?

A Bhutan



B Costa Rica

C New Zealand

D Norway

ANSWER & WHY

A. Bhutan

Bhutan formally adopted Gross National Happiness as a guiding development philosophy, weighing wellbeing, culture, environment, and governance alongside — and sometimes above — conventional economic growth.

When the Four Principles Pull Against Each Other

In the real world, these four principles rarely align perfectly — genuine sustainability work is about navigating the tension between them.

Resource Efficiency vs. Social Equity

Automation and efficiency gains can displace workers — efficiency for the planet can mean hardship for people, without a deliberate "just transition."

Economic Viability vs. Ecological Integrity

Conservation often has real short-term costs, while ecological damage is frequently cheaper in the near term — until the true cost is counted.

Social Equity vs. Economic Viability

Redistributive policies that improve equity can carry real short-term economic costs, testing political will.

Ecological Integrity vs. Resource Efficiency

The Jevons Paradox again: efficiency gains can enable more total consumption, undermining ecological goals unless paired with hard limits.

"A government proposes a large solar farm: it improves resource efficiency and long-term economic viability, but it will displace a farming community and requires clearing an intact grassland ecosystem. Using all four principles, how would you evaluate it?"

There is no single correct answer — notice which principle you instinctively weigh most heavily.

🕒 4 MIN

Four Principles, Recapped

1

Ecological Integrity

An ecosystem's composition, structure, and function — including its resilience to disturbance.

2

Planetary Boundaries

Six of nine boundaries already transgressed (Richardson et al., 2023) — the scientific evidence base for ecological limits.

3

Resource Efficiency

Achieving more value per unit of resource — the core goal is decoupling growth from environmental harm.

4

The IPAT Equation

Impact = Population × Affluence × Technology — a quantitative tool for reasoning about sustainability trade-offs.

5

The Jevons Paradox

Efficiency gains can increase total consumption rather than reduce it — a critical caveat, not a reason to abandon efficiency.

6

Social Equity

Fair distribution of both the benefits and burdens of development — distributive and procedural justice together.

7

Environmental Justice

From Warren County (1982) to Bhopal (1984): hazards are disproportionately sited near the least powerful communities.

8

Economic Viability

Activity that can genuinely continue — which requires internalizing externalities into true-cost pricing.

9

Beyond GDP

Green GDP, GPI, Bhutan's Gross National Happiness, and HDI all try to measure what GDP alone misses.

10

The Real Work

Genuine sustainability means navigating the honest tensions between all four principles — not pretending they never conflict.

The Sustainable Development Goals

All 17 goals, in full detail — and exactly how each one connects to climate change

Climate Action is not Goal 13 alone. It is threaded through all 17.

By the end of today, you will be able to explain the climate connection inside every single SDG — not just the one with "climate" in its name.

What Are the Sustainable Development Goals?

A single global to-do list, agreed to by every UN member state.

The Agenda

"Transforming our World: The 2030 Agenda for Sustainable Development" — adopted by all 193 UN member states on 25 September 2015, entering into force 1 January 2016.

The Structure

17 Goals, 169 targets, and over 230 indicators — covering poverty, health, education, climate, and everything in between, all due by 2030.

The Predecessor

The SDGs succeeded the Millennium Development Goals (2000–2015) — 8 goals that cut extreme poverty and child mortality sharply, but left environmental sustainability under-addressed.

The Promise

The Agenda's defining phrase: to "leave no one behind" — every goal is meant to apply universally, to rich and poor countries alike.

The Five Ps: How the 17 Goals Are Organized

The UN itself groups all 17 SDGs into five interconnected clusters — the structure we will follow today.



People

SDGs 1–5

End poverty and hunger, ensure health, education, and dignity for all.



Planet

SDGs 6, 12–15

Protect the planet from degradation for present and future generations.



Prosperity

SDGs 7–11

Ensure prosperous, fulfilling lives in harmony with nature.



Peace

SDG 16

Foster peaceful, just, and inclusive societies.



Partnership

SDG 17

Implement the agenda through a revitalized global partnership.

SECTION 1 OF 5

People

Ending poverty and hunger, ensuring health, education, and dignity — for every person, everywhere.



PEOPLE

1



No Poverty

End poverty in all its forms, everywhere.

KEY TARGETS

Eradicate extreme poverty for all people everywhere (living below \$2.15/day)

Ensure equal rights to economic resources, land, and basic services for the poor

Build the resilience of the poor to climate-related extreme events and disasters

CLIMATE CHANGE RELEVANCE

Climate change is a poverty multiplier. Droughts, floods, and crop failures hit the poorest households hardest — those with the fewest resources to recover — often pushing families that had escaped poverty back into it after a single climate shock.

World Bank projections: climate change could push tens of millions more people into extreme poverty by 2030 without urgent adaptation investment.



Zero Hunger

End hunger, achieve food security and improved nutrition, and promote sustainable agriculture.

KEY TARGETS

End hunger and ensure access to safe, sufficient, nutritious food for all, year-round

Ensure sustainable food production systems resilient to climate change and extreme weather

Maintain genetic diversity of seeds and crops to help adapt to changing conditions

CLIMATE CHANGE RELEVANCE

Changing rainfall patterns, heat stress, and extreme weather directly threaten crop yields and livestock. After decades of decline, global hunger has been rising again in recent years — with climate shocks now cited as one of the leading drivers.

FAO estimates hundreds of millions of people face chronic hunger worldwide, with climate volatility as a compounding cause.

PEOPLE

3



Good Health and Well-being

Ensure healthy lives and promote well-being for all, at all ages.

KEY TARGETS

Reduce premature mortality from non-communicable diseases through prevention

End epidemics of AIDS, tuberculosis, malaria, and other communicable diseases

Substantially reduce deaths from hazardous chemicals, air, water, and soil pollution

CLIMATE CHANGE RELEVANCE

Heat stress, expanding ranges of disease-carrying mosquitoes, worsening air quality, and food and water insecurity from climate change all directly increase disease burden. The WHO projects climate change will cause a substantial rise in additional deaths per year by mid-century.

WHO: climate change is projected to cause roughly 250,000 additional deaths per year between 2030–2050 from malnutrition, malaria, diarrhoea, and heat stress alone.

PEOPLE

4



Quality Education

Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.

KEY TARGETS

Ensure free, equitable, quality primary and secondary education for all children

Ensure learners acquire the knowledge needed to promote sustainable development, including climate change education (Target 4.7)

Build and upgrade education facilities that are safe, inclusive, and effective

CLIMATE CHANGE RELEVANCE

Floods, heatwaves, and storms destroy schools and disrupt learning for millions of children every year. Target 4.7 explicitly requires climate change education as part of quality education — meaning this very course is, in UN terms, a direct SDG 4.7 deliverable.

PEOPLE

5



Gender Equality

Achieve gender equality and empower all women and girls.

KEY TARGETS

End all forms of discrimination against women and girls everywhere

Ensure women's full and effective participation and equal leadership opportunities

Give women equal rights to economic resources, land ownership, and property

CLIMATE CHANGE RELEVANCE

Women and girls in developing countries are disproportionately burdened by climate impacts — walking further for water and fuel as droughts worsen — yet remain underrepresented in the climate decision-making that shapes their own futures.

Which "People" Goal Does This Climate Link Belong To?

Match each climate-related impact to the correct SDG in the People cluster.

		SDG 1–2	SDG 3–4	SDG 5
1	Crop failures from erratic rainfall drive hunger			
2	Heatwaves close schools and disrupt learning	1	2	5
3	Rising disease burden from expanding mosquito ranges	4	3	
4	Climate shocks push families back into extreme poverty		6	
5	Women walk further for water as droughts worsen			
6	Climate education is itself a stated global target			

SECTION 2 OF 5

Planet

Protecting the natural systems that make every other goal possible — the most directly climate-linked cluster.





Clean Water and Sanitation

Ensure availability and sustainable management of water and sanitation for all.

KEY TARGETS

Achieve universal, equitable access to safe and affordable drinking water

Substantially increase water-use efficiency and address water scarcity

Protect and restore water-related ecosystems, including rivers, wetlands, and lakes

CLIMATE CHANGE RELEVANCE

Climate change is intensifying droughts and shifting rainfall patterns, directly threatening water availability. Nearly half the world already faces severe water scarcity for at least one month a year — a figure set to worsen substantially as warming continues.

UN Water: roughly 2 billion people currently lack access to safely managed drinking water.

PLANET

12



Responsible Consumption and Production

Ensure sustainable consumption and production patterns.

KEY TARGETS

Achieve sustainable management and efficient use of natural resources

Halve global per-capita food waste at retail and consumer levels

Substantially reduce waste generation through prevention, reduction, recycling, and reuse

CLIMATE CHANGE RELEVANCE

How much we produce, consume, and waste drives resource extraction and emissions far upstream of the visible point of use. A more circular, less wasteful economy remains one of the most under-utilized levers for climate mitigation available today.



Climate Action

Take urgent action to combat climate change and its impacts.

KEY TARGETS

Strengthen resilience and adaptive capacity to climate-related hazards and disasters

Integrate climate change measures into national policies, strategies, and planning

Improve education, awareness, and institutional capacity on climate mitigation and adaptation

CLIMATE CHANGE RELEVANCE

This is the direct climate goal — and it explicitly recognizes the UNFCCC as the primary forum for international climate negotiation. Every topic in this course — footprint calculation, carbon markets, mitigation strategy — is, in UN terms, SDG 13 in practice.

SDG 13 is the connective tissue for all other 16 goals — nearly every one depends on climate stability to be achievable at all.

PLANET

14



Life Below Water

Conserve and sustainably use the oceans, seas, and marine resources.

KEY TARGETS

Prevent and significantly reduce marine pollution of all kinds

Sustainably manage and protect marine and coastal ecosystems

Minimize and address the impacts of ocean acidification

CLIMATE CHANGE RELEVANCE

Oceans have absorbed over 90% of the excess heat and roughly a quarter of the CO₂ from human emissions since the industrial era — warming and acidifying seas, bleaching coral reefs, and disrupting the fisheries that feed billions of people.

PLANET

15



Life on Land

Protect, restore, and promote sustainable use of terrestrial ecosystems, and halt biodiversity loss.

KEY TARGETS

- Conserve and restore terrestrial and inland freshwater ecosystems
- Halt deforestation, restore degraded forests, and increase afforestation
- Halt biodiversity loss and protect threatened species from extinction

CLIMATE CHANGE RELEVANCE

Forests and soils are among the planet's largest carbon sinks. Deforestation alone accounts for a significant share of global emissions — making land protection and restoration one of the most direct, immediately available climate mitigation tools we have.

Which SDG explicitly recognizes the UNFCCC as the primary international forum for climate negotiations?

- A SDG 6 (Clean Water and Sanitation)
- B SDG 12 (Responsible Consumption and Production)
- C SDG 13 (Climate Action)** ✓
- D SDG 15 (Life on Land)

ANSWER & WHY

C. SDG 13 (Climate Action)

SDG 13's targets explicitly reference the UNFCCC as the primary intergovernmental forum for negotiating the global response to climate change — making it the direct climate goal, distinct from the other Planet-cluster goals which address specific environmental systems.

SECTION 3 OF 5

Prosperity

Ensuring prosperous, fulfilling lives in harmony with nature — where the clean-energy transition lives.



PROSPERITY

7



Affordable and Clean Energy

Ensure access to affordable, reliable, sustainable, and modern energy for all.

KEY TARGETS

Ensure universal access to affordable, reliable, and modern energy services

Substantially increase the share of renewable energy in the global energy mix

Double the global rate of improvement in energy efficiency

CLIMATE CHANGE RELEVANCE

Energy production is the single largest source of global greenhouse gas emissions. SDG 7 is arguably the most directly climate-linked goal after SDG 13 itself, since the clean-energy transition is the primary lever available for global mitigation.

PROSPERITY

8



Decent Work and Economic Growth

Promote sustained, inclusive, and sustainable economic growth, full employment, and decent work for all.

KEY TARGETS

Improve resource efficiency in consumption and production, decoupling growth from environmental degradation

Achieve full and productive employment and decent work for all women and men

Protect labour rights and promote safe working environments for all workers

CLIMATE CHANGE RELEVANCE

The transition to a low-carbon economy is reshaping entire industries. Millions of "green jobs" are being created in renewables and climate tech, while fossil-fuel-dependent regions and workers require a deliberately managed "just transition."

PROSPERITY

9



Industry, Innovation and Infrastructure

Build resilient infrastructure, promote inclusive and sustainable industrialization, and foster innovation.

KEY TARGETS

Develop quality, reliable, sustainable, and resilient infrastructure

Upgrade industry and infrastructure for sustainability, with greater resource efficiency and cleaner technology

Enhance scientific research and upgrade the technological capabilities of industrial sectors

CLIMATE CHANGE RELEVANCE

Industry accounts for a substantial share of global emissions. SDG 9 directly drives the clean-technology innovation — efficient manufacturing, green infrastructure, low-carbon materials — that climate mitigation ultimately depends on at scale.

PROSPERITY

10



Reduced Inequalities

Reduce inequality within and among countries.

KEY TARGETS

Achieve income growth for the bottom 40% of the population faster than the national average

Empower and promote the social, economic, and political inclusion of all

Facilitate safe, orderly, and responsible migration and mobility of people

CLIMATE CHANGE RELEVANCE

Climate change hits the world's poorest and most vulnerable countries hardest, despite them having contributed the least to historical emissions — an injustice now central to international "loss and damage" climate finance negotiations.

PROSPERITY

11



Sustainable Cities and Communities

Make cities and human settlements inclusive, safe, resilient, and sustainable.

KEY TARGETS

Ensure access to adequate, safe, and affordable housing and basic services

Significantly reduce deaths and losses from disasters, including climate-related ones

Reduce the environmental impact of cities, including air quality and waste management

CLIMATE CHANGE RELEVANCE

Cities house over half of humanity and generate roughly 70% of global CO₂ emissions. Urban design, public transit, and building efficiency have become central climate battlegrounds — where mitigation and daily quality of life intersect directly.

Cities: ~56% of the world's population, but responsible for an estimated ~70% of global energy-related CO₂ emissions.

Direct or Indirect Climate Link?

For each Prosperity-cluster goal, decide whether its climate connection is direct (about emissions/mitigation) or indirect (climate as a compounding factor).

1	SDG 7 — Affordable and Clean Energy	DIRECT LINK	INDIRECT LINK
2	SDG 8 — Decent Work and Economic Growth	1	2
3	SDG 9 — Industry, Innovation and Infrastructure	3	4
4	SDG 10 — Reduced Inequalities		
5	SDG 11 — Sustainable Cities and Communities	5	

SECTION 4 OF 5

Peace

Strong, accountable institutions — the enforcement layer every other goal quietly depends on.



PEACE

16



Peace, Justice and Strong Institutions

Promote peaceful and inclusive societies, provide access to justice for all, and build effective, accountable institutions.

KEY TARGETS

Significantly reduce corruption and bribery in all their forms

Develop effective, accountable, and transparent institutions at all levels

Ensure responsive, inclusive, participatory, and representative decision-making

CLIMATE CHANGE RELEVANCE

Strong, transparent institutions are what make climate policy actually enforceable — from carbon market MRV to disaster response coordination. At the same time, climate-driven resource scarcity (water, farmland) is an increasing driver of conflict and instability.

SECTION 5 OF 5

Partnership

No single country — or company, or student — can deliver this agenda alone.



PARTNERSHIP

17



Partnerships for the Goals

Strengthen the means of implementation and revitalize the global partnership for sustainable development.

KEY TARGETS

Enhance the global partnership for sustainable development, complemented by multi-stakeholder partnerships

Encourage effective public, public-private, and civil-society partnerships

Develop better measurements of progress on sustainable development beyond GDP alone

CLIMATE CHANGE RELEVANCE

No single country can solve climate change alone. SDG 17 underpins every climate agreement covered in this course — the Paris Agreement, Article 6 carbon trading, and international climate finance transfers are all, fundamentally, SDG 17 in motion.

"Peace and Partnership rarely make climate headlines — but which recent climate story have you seen that actually depended on one of them working?"

Think about disaster response, cross-border agreements, or corruption in climate finance.

🕒 3 MIN

BRINGING IT TOGETHER

Climate Change Threads Through All 17 Goals

Not every goal is about climate — but every goal is affected by it, or affects it in return.

1–5 (People)

Climate change is a threat multiplier — worsening poverty, hunger, disease, and inequality for the most vulnerable first.

6, 12, 14, 15 (Planet)

The natural systems most directly damaged by, and most protective against, a warming climate.

13 (Planet)

The direct goal — climate mitigation and adaptation as an explicit global target in its own right.

7–11 (Prosperity)

The economic engine of the solution — clean energy, green jobs, and sustainable industry and cities.

16 (Peace)

The institutional capacity that makes climate policy and disaster response actually enforceable.

17 (Partnership)

The cooperative machinery — finance, treaties, and technology transfer — that makes global action possible at all.

Most Direct Climate Link, or Most Indirect?

Across all 17 goals, which of these have the most direct climate connection versus the most indirect (but still real) one?

		MOST DIRECT	MORE INDIRECT
1	SDG 13 — Climate Action		
2	SDG 7 — Affordable and Clean Energy	1	3
3	SDG 5 — Gender Equality	2	5
4	SDG 15 — Life on Land	4	6
5	SDG 16 — Peace, Justice and Strong Institutions		
6	SDG 4 — Quality Education		

"Of all 17 goals, which one would you personally want to contribute to through your own engineering career — and why?"

There is no single right answer. Notice which goal your mind goes to first.

🕒 3 MIN

The 17 Goals, Recapped

1

The Agenda

193 countries, 17 goals, 169 targets, adopted 2015 — the world's shared to-do list through 2030.

2

People (1–5)

Poverty, hunger, health, education, and gender equality — all worsened by climate shocks.

3

Planet (6, 12–15)

Water, consumption, climate action itself, oceans, and land — the environmental foundation.

4

SDG 13 Specifically

The direct climate goal, recognizing the UNFCCC as the primary negotiating forum.

5

Prosperity (7–11)

Clean energy, decent work, industry, equality, and cities — the economic engine of the solution.

6

Peace (16)

Strong institutions make climate policy enforceable; climate stress can itself drive conflict.

7

Partnership (17)

No single country solves this alone — every climate treaty in this course depends on SDG 17.

8

Direct vs. Indirect Links

Some goals (7, 13, 15) are directly about climate; others (4, 5, 16) are linked through compounding effects.

9

The Core Insight

Climate change is not one goal among seventeen — it is a thread running through all of them.

10

Your Role

Every goal has an engineering gap waiting to be filled — the question is only which one calls to you.

Carbon Footprint Calculation

Scope 1, Scope 2 & Scope 3 emissions — definitions, formulas, and fully worked calculations under the GHG Protocol

**"Net zero by 2070" is a sentence.
A carbon footprint calculation is the arithmetic that
makes it real.**

Every climate pledge you've heard about — corporate, national, personal — reduces to the same three numbers you will learn to calculate today.

From Definitions to a Full Calculation



01

Foundations of Carbon Accounting



02

The GHG Protocol Framework



03

Scope 1 — Direct Emissions



04

Scope 2 — Energy Indirect Emissions



05

Scope 3 — Value Chain Emissions



06

Full Worked Case Study



07

Tools, Data & the Indian Context

SECTION 1 OF 7

Foundations of Carbon Accounting

Before any formula: what exactly are we counting, and in what units?



What Is a Carbon Footprint?

A precise, measurable quantity — not a metaphor.

Working Definition

The total greenhouse gas emissions caused directly and indirectly by an organization, product, event, or individual, expressed in a single common unit.

Why "Carbon" When We Mean Several Gases

CO₂, methane, nitrous oxide and others all trap heat differently — "carbon footprint" is shorthand for the combined warming effect of all of them together.

The Unit That Makes This Possible

tCO₂e — tonnes of carbon dioxide equivalent. Every gas is converted into "how much CO₂ would cause the same warming," so they can be added together.

Global Warming Potential (GWP): Converting Gases to CO₂e

1

Carbon Dioxide (CO₂) — the reference gas

27–30

Methane (CH₄) — over a 100-year period

273

Nitrous Oxide (N₂O) — over a 100-year period

25,200

Sulfur Hexafluoride (SF₆) — over a 100-year period

Formula: Mass of Gas (tonnes) × GWP = tCO₂e. This is what lets a factory's methane leak and a car's CO₂ exhaust sit on the same balance sheet.

Source: IPCC AR6 Working Group I (2021), 100-year Global Warming Potential values.

Why Organizations Measure Their Footprint

Measurement is now a regulatory, financial, and strategic requirement — not a goodwill gesture.

Regulatory Compliance

India's SEBI mandates BRSR (Business Responsibility and Sustainability Reporting) disclosure for the top 1,000 listed companies by market cap, including Scope 1 and 2 emissions.

Carbon Markets

India's Carbon Credit Trading Scheme (CCTS), run by the Bureau of Energy Efficiency, prices emissions directly — you cannot trade what you cannot measure.

Investor & Customer Pressure

Frameworks like CDP and the Science Based Targets initiative (SBTi) require verified Scope 1, 2, and increasingly Scope 3 data before certifying any climate target.

You Can't Manage What You Don't Measure

Every reduction strategy — efficiency, renewables, supply-chain redesign — starts from a credible baseline number.

A company emits 2 tonnes of methane (CH_4) in a year. Using a GWP of 28, how many tCO_2e does this represent?

A 2 tCO_2e

B 28 tCO_2e

C 56 tCO_2e



D 280 tCO_2e

ANSWER & WHY

C. 56 tCO_2e

Mass of gas $\times$ GWP = tCO_2e . Here: 2 tonnes $\times$ 28 = 56 tCO_2e . This is the exact calculation used to fold non- CO_2 gases into a single footprint number.

SECTION 2 OF 7

The GHG Protocol Framework

The global standard behind every "Scope 1/2/3" number you will ever see reported.

The GHG Protocol: The Global Standard

Developed jointly by the World Resources Institute (WRI) and the World Business Council for Sustainable Development (WBCSD).

What It Is

The most widely used international accounting framework for measuring and reporting greenhouse gas emissions — the source of the "Scope 1/2/3" terminology used worldwide.

Who Uses It

The overwhelming majority of Fortune 500 companies reporting emissions use the GHG Protocol as their accounting foundation, alongside frameworks like ISO 14064-1 which align closely with it.

Why It Matters For This Course

Every calculation in this lecture — and every case study in this course — is built on this exact framework.

Setting Organizational Boundaries



Equity Share Approach

Account for emissions in proportion to your % ownership of each operation.

A 40%-owned joint venture contributes 40% of its emissions to your inventory.

Common in investment and holding-company contexts.



Control Approach (Financial or Operational)

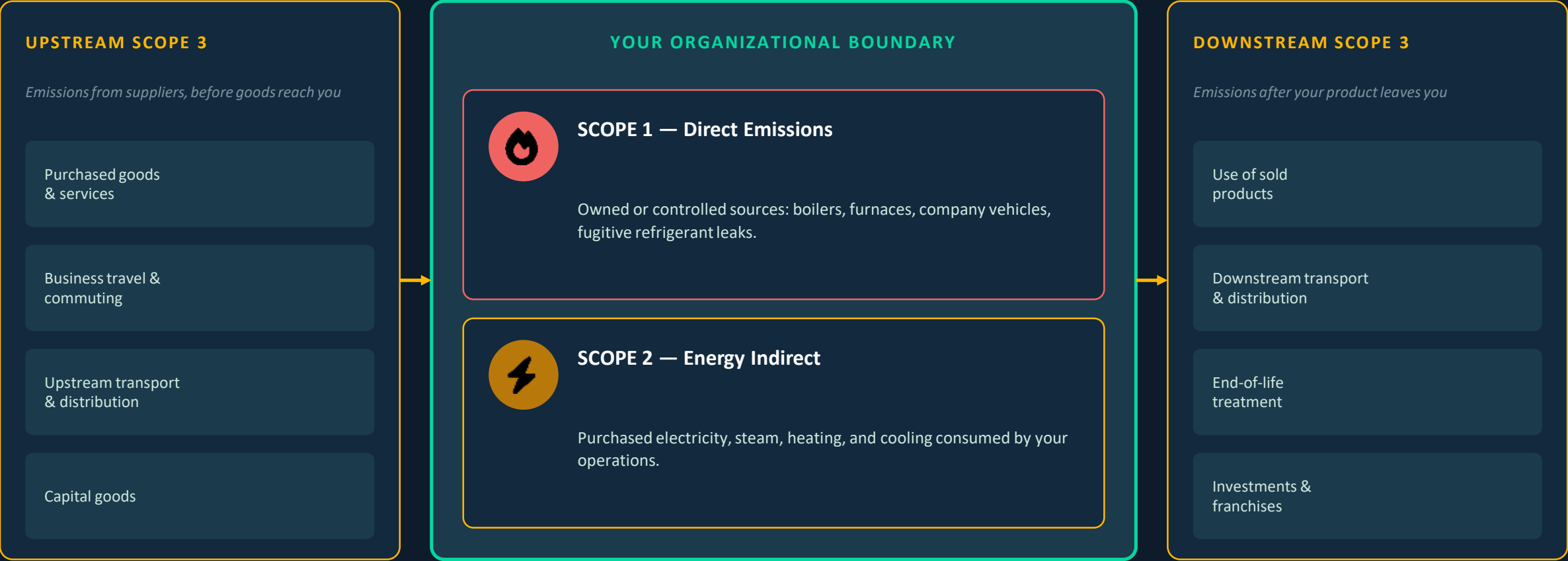
Account for 100% of emissions from operations you control — regardless of ownership %.

Financial control: you direct the entity's financial policy. Operational control: you direct its operating policy.

The most common approach for corporate operational reporting — including BRSR in India.

The Three Scopes, At a Glance

Every emission source in the economy falls into exactly one of these three boxes, relative to your organization.



Scope 1 vs. 2 vs. 3 — Side by Side

 Scope 1 & 2	 Scope 3
Directly measurable from your own meters, fuel receipts, and utility bills.	Estimated from supplier, customer, and market-average data — rarely directly metered.
Mandatory under nearly every reporting framework, including BRSR.	Increasingly required (CSRD, SBTi) but historically voluntary in many jurisdictions.
Typically 10–30% of a company's total footprint.	Typically 70–90% of a company's total footprint, across most industries.

Which Scope Does This Belong To?

For each emission source, decide which scope it falls under.

1	Diesel burned in a company-owned generator	SCOPE 1	SCOPE 2	SCOPE 3
2	Electricity purchased from the grid to run office lighting	1	2	3
3	Emissions from a supplier manufacturing your raw materials	4	6	5
4	Fuel burned in company-owned delivery trucks			
5	Employees commuting to work in their own cars			
6	Purchased steam used in a factory process			

SECTION 3 OF 7

Scope 1 — Direct Emissions

Emissions from sources you own or control — the most direct entry in the ledger.



Scope 1 — Direct Emissions

GHG emissions from sources owned or controlled by the reporting organization, falling into four categories.



Stationary Combustion

Fuel burned in boilers, furnaces, and generators at fixed facilities.



Mobile Combustion

Fuel burned in company-owned or -leased vehicles — cars, trucks, forklifts.



Fugitive Emissions

Unintentional releases — refrigerant (HFC) leaks from AC and cooling systems, equipment leaks.



Process Emissions

Released directly by an industrial process itself, not combustion — e.g. cement calcination, chemical manufacturing.

The Universal Emissions Calculation Formula

This same structure — with different activity data and factors — calculates Scope 1, 2, and most of Scope 3.

$$\text{Emissions (tCO}_2\text{e)} = \text{Activity Data} \times \text{Emission Factor} \times \text{GWP}$$

Activity Data

The quantity of a fuel, material, or service consumed — litres of diesel, kWh of electricity, tonnes of steel, km travelled.

Emission Factor

A published, standardized figure converting one unit of activity into GHG mass — e.g. kgCO₂e per litre of diesel, sourced from IPCC, DEFRA, EPA, or CEA.

GWP

Applied only when the emission factor is not already expressed in CO₂e — converts a non-CO₂ gas mass into its CO₂-equivalent warming effect.

SECTION 3 · WORKED EXAMPLE

Scope 1: Emissions From Fuel Combustion

GIVEN

- Diesel consumed by backup generators: 5,000 litres/year
- Petrol consumed by company vehicles: 2,000 litres/year
- Emission factor, diesel (typical published factor): 2.68 kgCO₂e/litre
- Emission factor, petrol (typical published factor): 2.31 kgCO₂e/litre

FORMULA

$$\text{Scope 1} = \Sigma (\text{Fuel Volume} \times \text{Emission Factor}) \div 1000$$

STEP-BY-STEP CALCULATION

- 1 Diesel: 5,000 L × 2.68 kgCO₂e/L = 13,400 kgCO₂e
- 2 Petrol: 2,000 L × 2.31 kgCO₂e/L = 4,620 kgCO₂e
- 3 Sum the two sources: 13,400 + 4,620 = 18,020 kgCO₂e
- 4 Convert kg to tonnes: 18,020 ÷ 1,000 = 18.02 tCO₂e

18.02 tCO₂e

Total Scope 1 emissions for the year, from stationary generator combustion plus mobile vehicle combustion combined.

Scope 1: Your Turn

"A campus canteen uses an LPG cylinder-fed cooking system, consuming 600 kg of LPG per year. What are its annual Scope 1 emissions?"

GIVEN: LPG consumption = 600 kg/year; Emission factor for LPG (typical published factor) = 2.98 kgCO₂e/kg

SUGGESTED SOLUTION

Scope 1 = Activity Data × Emission Factor

- $600 \text{ kg} \times 2.98 \text{ kgCO}_2\text{e/kg} = 1,788 \text{ kgCO}_2\text{e}$
- Convert to tonnes: $1,788 \div 1,000 = 1.788 \text{ tCO}_2\text{e}$

ANSWER

$\approx 1.79 \text{ tCO}_2\text{e per year}$

SECTION 4 OF 7

Scope 2 — Energy Indirect Emissions

Emissions from the energy you buy — generated elsewhere, but caused by your consumption.



Scope 2 — Purchased Energy

Indirect emissions from the generation of electricity, steam, heating, or cooling that your organization purchases and consumes.

Why "Indirect"

The physical combustion happens at the power plant, not on your premises — but your consumption decision causes it.

What Counts

Purchased grid electricity is the most common source; purchased steam, heat, and chilled water for cooling also fall under Scope 2.

Why It Gets Its Own Scope

Purchased energy is significant enough, and different enough from on-site combustion, that the GHG Protocol treats it as a distinct category with its own calculation methods.

Location-Based vs. Market-Based Method



Location-Based Method

Uses the average emission intensity of the grid(s) where consumption physically occurs.

In India: the CEA's published grid-average emission factor.

Answers: "How clean is the grid I'm drawing from?"

The same for every company on the same grid.



Market-Based Method

Uses emissions specific to the electricity contracts an organization has actually purchased.

Reflects renewable energy certificates (RECs), power purchase agreements (PPAs), and green tariffs.

Answers: "How clean is the electricity I chose to buy?"

Can fall close to zero if 100% renewable energy is contractually purchased.

SECTION 4 · WORKED EXAMPLE

Scope 2: Using India's Grid Emission Factor

GIVEN

- Annual grid electricity consumption: 500,000 kWh (500 MWh)
- India weighted average grid emission factor (CEA CO₂ Baseline Database, FY 2023–24): 0.727 tCO₂/MWh
- Method used: Location-based

FORMULA

$$\text{Scope 2 (location-based)} = \text{Electricity Consumed (MWh)} \times \text{Grid Emission Factor (tCO}_2\text{/MWh)}$$

STEP-BY-STEP CALCULATION

- 1 Convert consumption to MWh: 500,000 kWh = 500 MWh
- 2 Apply the CEA grid emission factor: 500 MWh × 0.727 tCO₂/MWh = 363.5 tCO₂e
- 3 Note: if this energy were instead 100% covered by a renewable PPA, market-based Scope 2 could approach 0 tCO₂e — which is exactly why the GHG Protocol requires reporting both methods side by side.

363.5 tCO₂e

Location-based Scope 2 emissions for the year, using the Central Electricity Authority's published grid-average emission factor.

Scope 2: Your Turn

"A data center consumes 1,200 MWh of grid electricity annually. Calculate its location-based Scope 2 emissions."

GIVEN: Electricity consumption = 1,200 MWh/year; CEA FY 2023–24 grid emission factor = 0.727 tCO₂/MWh

SUGGESTED SOLUTION

Scope 2 = Electricity Consumed (MWh) × Grid Emission Factor

- 1,200 MWh × 0.727 tCO₂/MWh = 872.4 tCO₂e

ANSWER

872.4 tCO₂e per year

SECTION 5 OF 7

Scope 3 — Value Chain Emissions

Everything else in the value chain — usually the hardest to measure, and usually the largest.



Scope 3 — Value Chain Emissions

All indirect emissions that occur in an organization's value chain, excluding Scope 2 — both upstream (suppliers) and downstream (customers).

Not Owned, Not Controlled

These emissions physically happen at other companies' facilities, in other people's vehicles, or inside products after they've been sold.

Why It's Hardest to Measure

Reliable data means asking suppliers and customers for their own emissions — data an organization rarely has direct access to.

Why It's Still Required

For most industries, Scope 3 is 70–90% of the total footprint. Skipping it means missing most of the real number.

Upstream Scope 3 — Categories 1 to 8

Everything that happens before a product or service reaches your organization.



1. Purchased Goods & Services

Emissions from producing everything your organization buys or acquires.



2. Capital Goods

Emissions from producing purchased long-term assets — machinery, buildings, equipment.



3. Fuel- & Energy-Related Activities

Upstream emissions of fuel and energy not already counted in Scope 1 or 2.



4. Upstream Transportation & Distribution

Third-party transport of purchased products to your organization.



5. Waste Generated in Operations

Emissions from third-party disposal and treatment of your operational waste.



6. Business Travel

Employee travel for business in vehicles not owned by the organization — flights, trains, taxis.



7. Employee Commuting

Emissions from employees' travel between home and work.



8. Upstream Leased Assets

Emissions from operating assets you lease, not already in Scope 1 or 2.

Downstream Scope 3 — Categories 9 to 15

Everything that happens after your product or service leaves your organization.



9. Downstream Transportation

Transport and distribution of sold products, after they leave your control.



10. Processing of Sold Products

Emissions from further processing of your products by downstream companies.



11. Use of Sold Products

Emissions from customers actually using the products you sold — often the largest downstream category.



12. End-of-Life Treatment

Emissions from disposal and treatment of sold products at the end of their life.



13. Downstream Leased Assets

Emissions from assets you own but lease out to others to operate.



14. Franchises

Emissions from the operation of franchises not already counted elsewhere.



15. Investments

Emissions financed through equity, debt, and project investments — largest for banks and asset managers.

Four Ways to Calculate Scope 3

Data availability, not preference, usually decides which method you use — and different categories often need different methods.

Supplier-Specific Method

Uses actual emissions data reported by your own suppliers. Most accurate — but requires supplier cooperation few companies fully have yet.

Hybrid Method

Combines supplier-specific data where available with secondary (industry-average) data to fill the gaps.

Average-Data Method

Activity data (mass, distance, count) × an industry-average emission factor — e.g. tonnes of steel × tCO₂e per tonne of steel.

Spend-Based Method

Money spent × an economic input-output (EEIO) emission factor. Fastest and most available — but the least precise; often used as a first screening pass.

Scope 3, Category 1: Average-Data Method

GIVEN

- Steel purchased for manufacturing: 200 tonnes/year
- Industry-average emission factor for crude steel production (e.g. worldsteel Association): ≈ 1.9 tCO₂e per tonne
- Method used: Average-data method

FORMULA

$$\text{Category 1 Emissions} = \text{Mass of Material Purchased (tonnes)} \times \text{Material Emission Factor (tCO}_2\text{e/tonne)}$$

STEP-BY-STEP CALCULATION

- 1 Apply the factor: $200 \text{ tonnes} \times 1.9 \text{ tCO}_2\text{e/tonne} = 380 \text{ tCO}_2\text{e}$
- 2 This covers steel alone — a complete Category 1 figure sums this calculation across every material, component, and service purchased in the year.

380 tCO₂e

Category 1 (Purchased Goods & Services) emissions attributable to steel procurement for the year.

Upstream or Downstream Scope 3?

For each Scope 3 category, decide whether it is upstream (before the sale) or downstream (after the sale).

	UPSTREAM	DOWNSTREAM
1 Business Travel	1	2
2 Use of Sold Products	3	4
3 Employee Commuting	5	6
4 End-of-Life Treatment of Sold Products		
5 Purchased Goods & Services		
6 Investments		

SECTION 6 OF 7

Full Worked Case Study

Putting all three scopes together — one company, one full-year inventory.



Case Study: GreenTech Manufacturing Pvt. Ltd.

An illustrative mid-size electronics-component manufacturer, built to walk through a complete, realistic Scope 1–2–3 inventory.



Location

Pune, Maharashtra — one manufacturing facility plus a corporate office



Workforce

450 employees across manufacturing and administrative functions



Sector

Electronics component manufacturing for industrial and consumer clients



Reporting Year

Full financial year, FY 2023–24 — aligned to the CEA grid factor used

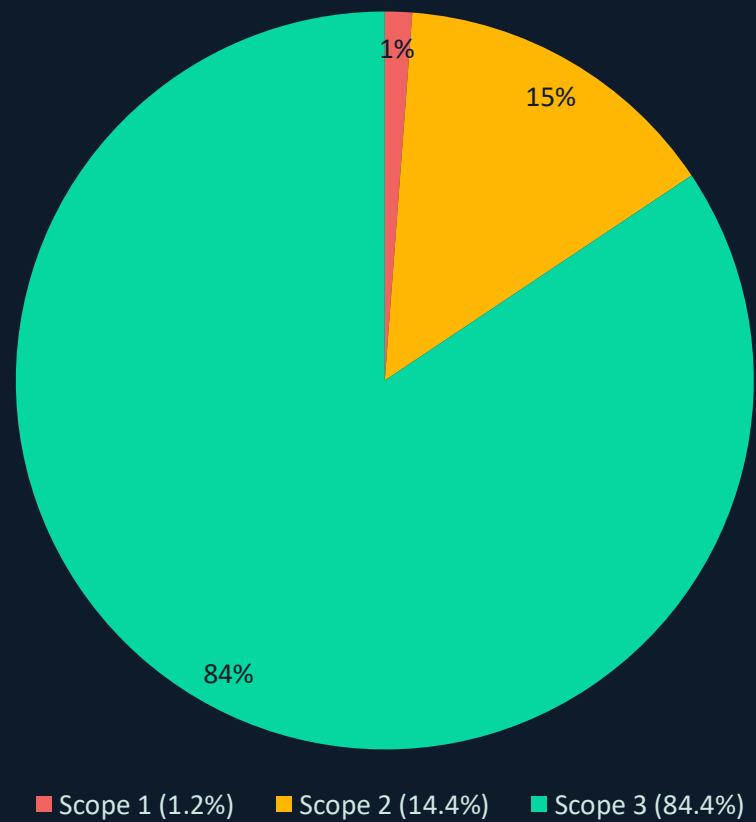
Every figure that follows is illustrative — built for this lecture — but every method, factor type, and proportion mirrors real published patterns.

The Complete Calculation, Line by Line

SCOPE	LINE ITEM	tCO ₂ e
SCOPE 1	Diesel generators + company vehicles	120
SCOPE 2	Grid electricity — 2,000 MWh × 0.727 tCO ₂ /MWh	1,454
SCOPE 3	Cat. 1 — Purchased goods & materials	5,200
SCOPE 3	Cat. 4 — Upstream transportation	620
SCOPE 3	Cat. 6 & 7 — Business travel & commuting	380
SCOPE 3	Cat. 11 — Use of sold products	2,100
SCOPE 3	All other categories (waste, end-of-life, etc.)	200
TOTAL ANNUAL FOOTPRINT		10,074 tCO ₂ e

Visualizing the Footprint

10,074 tCO₂e total — split across all three scopes.



THE LESSON

- Scope 1 — the easiest to measure — is only 1.2% of the real footprint.
- Scope 2 is manageable and directly reducible through renewable power purchase.
- Scope 3 is 84% of the total, driven overwhelmingly by purchased materials and product use.
- A company that reports only Scope 1 and 2 is reporting roughly one-sixth of its true footprint.

"Given this breakdown, where should GreenTech focus its reduction efforts first — and why might that be harder than it looks?"

Consider: what's easiest to act on directly, versus what actually moves the total number.

🕒 3 MIN

SECTION 7 OF 7

Tools, Data & the Indian Context

Where the emission factors come from, who requires this data, and what automates it at scale.



Where Emission Factors Actually Come From

Every calculation in this lecture depended on a published, standardized emission factor — these are the primary sources.

IPCC

The Intergovernmental Panel on Climate Change publishes default global emission factors and GWP values used as the international baseline.

DEFRA (UK)

The UK Department for Environment, Food & Rural Affairs publishes an annually updated set of GHG conversion factors, widely used worldwide as reference data — including the fuel factors used in this lecture.

US EPA

The US Environmental Protection Agency's GHG Emission Factors Hub covers fuels, electricity, and materials for the US context.

CEA (India)

India's Central Electricity Authority publishes the annual CO₂ Baseline Database for the Indian Power Sector — the authoritative source for India's grid emission factor.

Reporting Frameworks Built On These Numbers

The calculation methodology is identical — what changes is who you report it to, and how it gets verified.

CDP (Carbon Disclosure Project)

A global voluntary disclosure platform used by investors and customers to assess corporate climate risk and performance.

SBTi (Science Based Targets initiative)

Requires verified Scope 1, 2, and (increasingly) Scope 3 data before certifying that a company's emissions-reduction target is scientifically credible.

BRSR (India)

SEBI's Business Responsibility and Sustainability Reporting mandate for India's top 1,000 listed companies by market cap — Scope 1 and 2 disclosure required, Scope 3 increasingly expected.

CCTS (India)

India's Carbon Credit Trading Scheme, administered by the Bureau of Energy Efficiency, prices emissions directly using CEA grid data as its baseline reference.

What Automates This At Scale

The hand calculations you just did are exactly what carbon accounting software automates across thousands of line items.

Carbon Accounting Platforms

Tools like Watershed and Persefoni ingest utility bills, spend data, and supplier records to run this same formula automatically, continuously.

Emission Factor APIs

Platforms like Climatiq expose emission factor databases as an API, letting developers query "diesel, India, 2024" and get back a verified factor programmatically.

The Engineering Problem Underneath

Every platform like this is, at its core, a data pipeline problem: ingest messy activity data, match it to the right emission factor, apply the right GWP, and produce an auditable number.

Which Indian regulatory body publishes the annual grid emission factor used in Scope 2 calculations?

A SEBI (Securities and Exchange Board of India)

B BEE (Bureau of Energy Efficiency)

C CEA (Central Electricity Authority) ✓

D MoEFCC (Ministry of Environment, Forest and Climate Change)

ANSWER & WHY

C. CEA (Central Electricity Authority)

The Central Electricity Authority (CEA) publishes the annual CO₂ Baseline Database for the Indian Power Sector, which is the authoritative source for India's grid emission factor — used directly in the Scope 2 worked example earlier in this lecture.

The Complete Calculation, Recapped

1

Carbon Footprint

Total GHG emissions from an entity, expressed in tCO₂e using GWP to combine different gases.

2

The GHG Protocol

The global standard (WRI/WBCSD) behind every Scope 1/2/3 number reported worldwide.

3

Organizational Boundaries

Set first, using either the equity-share or control approach, before any calculation begins.

4

Scope 1

Direct emissions from owned or controlled sources — stationary, mobile, fugitive, and process.

5

Scope 2

Indirect emissions from purchased energy — calculated via location-based or market-based methods.

6

Scope 3

15 categories of value-chain emissions, upstream and downstream — usually 70–90% of the total.

7

The Universal Formula

Emissions = Activity Data × Emission Factor × GWP — the same structure across all three scopes.

8

The Case Study Lesson

A company reporting only Scope 1 and 2 may be capturing roughly one-sixth of its real footprint.

9

India's Framework

CEA grid factors, BRSR mandatory disclosure, and the CCTS carbon market all build on this exact methodology.

10

Your Opportunity

Every one of these calculations is a data pipeline problem waiting for better software.

Carbon Trading, Pricing & Risk Management

How a tonne of CO₂ gets a price, how that price is traded, and what can go wrong along the way

**The same tonne of CO₂ can be worth
€72 in Europe, \$5 on the voluntary market, or \$190 to
society.**

That price gap is not a rounding error — it is the entire subject of today's lecture, and it is where fortunes, frauds, and genuine climate progress all happen at once.

From "Why a Market?" to "What Could Go Wrong?"



01

Why Carbon Needs a Market



02

Carbon Pricing Mechanisms



03

Compliance Markets (Cap-and-Trade)



04

Voluntary Markets & Offsets



05

India's Carbon Market — CCTS



06

Risk Management in Carbon Markets



07

The Road Ahead

SECTION 1 OF 7

Why Carbon Needs a Market

The economic logic that turns an invisible gas into a priced, tradable commodity.



The Externality Problem

A factory that burns coal doesn't pay for the warming it causes — a third party (the whole planet) bears that cost instead. Economists call this a negative externality.

The Market Failure

Without intervention, the price of fossil energy reflects only extraction and production cost — never the climate damage it causes downstream.

The Economic Fix

Put an explicit price on emissions, so the cost of climate damage is "internalized" into the decisions of the emitter — exactly what a market failure requires.

Two Ways To Do It

Governments have converged on two dominant tools to set that price: taxing emissions directly, or capping total emissions and letting a market set the price.

Carbon Tax vs. Cap-and-Trade



Carbon Tax

Government sets the price directly; the market determines the resulting quantity of emissions.

Simple to administer, highly predictable cost for businesses.

Emissions reduction is uncertain — it depends how businesses respond to the price.

Examples: Sweden, Canada's federal backstop, Singapore.



Cap-and-Trade (ETS)

Government sets the total quantity (the cap) directly; the market determines the resulting price.

Guarantees a specific emissions outcome — the cap is a hard ceiling.

Price is uncertain and can be volatile, since it depends on market supply and demand.

Examples: EU ETS, California Cap-and-Trade, China National ETS, India's CCTS.

Under a cap-and-trade system, what does the government set directly?

- A The exact price of carbon
- B The total quantity of allowed emissions (the cap) ✓**
- C Which companies must participate
- D The exchange rate between currencies

ANSWER & WHY

B. The total quantity of allowed emissions (the cap)

Cap-and-trade fixes the quantity (the cap) and lets the market discover the price through trading. A carbon tax does the reverse — it fixes the price and lets the market determine the resulting quantity of emissions.

SECTION 2 OF 7

Carbon Pricing Mechanisms

Once you decide to price carbon, the next question is: what price?



What Actually Sets a Carbon Price?

Unlike a carbon tax, market-based carbon prices emerge the same way any commodity price does.

Auctions

Regulators periodically auction newly issued allowances — bidders compete on price, setting a fresh market signal.

Secondary Trading

Once issued, allowances and credits trade continuously between companies, banks, and traders on exchanges like ICE and EEX.

Supply & Demand Fundamentals

Economic activity, fuel prices, weather (affecting renewable output), and policy tightening all shift the price, exactly like any other commodity.

The Social Cost of Carbon (SCC)

\$51

US interim federal estimate (2021), based on older methodology

\$190

US EPA's updated central estimate (2023), reflecting newer climate-damage science

~4×

How much the estimate grew in two years — a reminder that this number is genuinely contested

The SCC estimates the true economic damage of one extra tonne of CO₂ — it is a policy benchmark, not a traded price, and it remains one of the most politically contested numbers in climate economics.

Source: US EPA Report on the Social Cost of Greenhouse Gases (Nov 2023); Resources for the Future.

Carbon Prices Around the World, Compared

€70–75

EU ETS allowance price (2026) — the world's largest compliance market

\$4–6

Typical voluntary carbon credit price (2024 average asking price)

\$190

The EPA's estimate of what a tonne of CO₂ actually costs society

The same tonne of CO₂ is priced completely differently depending on which market it passes through — and closing that gap is one of the defining challenges of carbon market design.

Source: EU ETS (ICE/EEX, 2026); Carboncredits.com voluntary market data (2024); US EPA SCC (2023).

"If the true cost to society is around \$190/tonne, but you can buy a voluntary credit for \$5, what does that gap tell you about the credit's quality — or about the price?"

Hold this thought — Section 4 will test it directly.

🕒 2 MIN

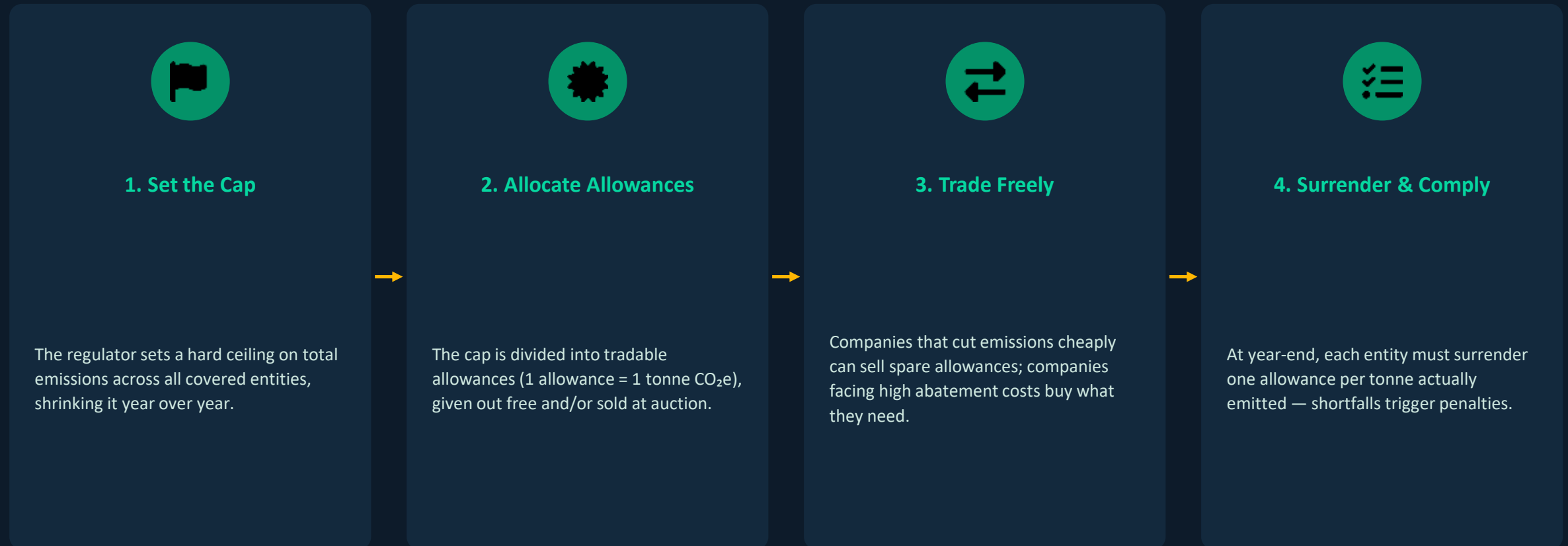
SECTION 3 OF 7

Compliance Markets (Cap-and-Trade)

The world's largest, most regulated carbon markets — where participation is mandatory, not optional.

How Cap-and-Trade Actually Works

Four steps turn a policy goal into a functioning market.



Free Allocation vs. Auctioning



Free Allocation

Allowances given to companies at no cost, based on historical output or benchmarks.

Protects industries exposed to international competition from "carbon leakage."

Risk: can hand out windfall profits if allowances are given generously.



Auctioning

Allowances sold to the highest bidder — companies pay the full market price upfront.

Generates government revenue that can fund climate programs (as EU ETS auction revenue does).

Creates a stronger, cleaner price signal from day one — no ambiguity about opportunity cost.

The EU ETS: The World's First and Largest Carbon Market

2005

Year the EU ETS launched — the world's first major carbon market

€70–75

Current allowance (EUA) price per tonne (2026)

4.3–4.4%

Annual rate the cap shrinks (the Linear Reduction Factor), tightening supply every year

2026

Year the Carbon Border Adjustment Mechanism (CBAM) enters full enforcement on imports

CBAM extends the EU's carbon price to imported steel, cement, and aluminium — so non-EU producers can no longer undercut EU producers on carbon cost alone.

Source: European Commission; ICE/EEX carbon price data (2026); EU CBAM regulation.

Calculating a Compliance Position

GIVEN

- A cement plant emits 80,000 tCO₂e this year
- It was allocated 65,000 free allowances
- Current EU ETS allowance (EUA) price: €72/tonne

FORMULA

$$\begin{aligned}\text{Allowance Shortfall} &= \text{Emissions} - \text{Allowances Held} \\ \text{Compliance Cost} &= \text{Shortfall} \times \text{Market Price}\end{aligned}$$

STEP-BY-STEP CALCULATION

- 1 Shortfall: $80,000 - 65,000 = 15,000$ allowances short
- 2 Cost to buy the shortfall: $15,000 \times €72 = €1,080,000$
- 3 This is the direct financial incentive to cut emissions rather than simply buy allowances.

€1,080,000

The plant's cost to purchase enough EUAs on the market to legally cover its emissions shortfall for the year.

Cap-and-Trade Term or Not?

For each term, decide whether it is a core cap-and-trade mechanism or belongs to a different pricing tool.

		CAP-AND-TRADE	CARBON TAX
1	A fixed fee per tonne of CO ₂ emitted		
2	A shrinking annual emissions ceiling	2	1
3	Tradable allowances bought and sold between companies	3	4
4	Government sets the price; the market sets the quantity		
5	A Linear Reduction Factor tightening supply each year	5	
6	Predictable cost, uncertain emissions outcome		6

SECTION 4 OF 7

Voluntary Markets & Offsets

Where anyone — company or individual — can buy a claim to emissions reduced elsewhere.



Carbon Allowances vs. Carbon Credits



Allowances (Compliance)

A permission slip: "the right to emit one tonne," issued under a legal cap.

Total supply is fixed by regulation — scarcity is enforced by law.

Value comes from legal scarcity within a mandatory scheme.

Traded within EU ETS, CCTS, California Cap-and-Trade, etc.



Credits (Offsets)

A claim: "one tonne was actually reduced or removed elsewhere," from a specific project.

Supply grows whenever a new project is registered and verified — no legal cap.

Value depends entirely on whether the underlying project's claim is actually true.

Traded on the voluntary carbon market (VCM) — Verra, Gold Standard, and others.

What Makes a Carbon Credit Credible?

A tonne claimed on paper is worthless unless it clears four tests.



Additionality

The reduction would not have happened anyway, without the carbon credit revenue as an incentive.



Permanence

The carbon stays out of the atmosphere for the long term — not released again by fire, logging, or reversal.



Verification

An independent third party confirms the claimed reduction actually occurred, using agreed methodology.



No Double-Counting

The same tonne of reduction is claimed by only one buyer, in only one registry, exactly once.

Who Runs the Voluntary Carbon Market?

Independent, non-governmental standards bodies certify most voluntary credits — there is no single global regulator.

Verra (VCS)

The largest voluntary standard, having issued more than 1 billion carbon credits since 2009 across renewable energy, forestry, and other project types.

Gold Standard

Founded with WWF backing, known for stricter co-benefit requirements — notably declining to certify credits based on avoided deforestation alone.

ACR & CAR

The American Carbon Registry and Climate Action Reserve — North America-focused standards with their own methodologies and registries.

When "Verified" Credits Turned Out to Be Phantom



The Guardian / Die Zeit / SourceMaterial Investigation (Jan 2023)

A nine-month investigation found that over 90% of Verra's rainforest (REDD+) offset credits — the most commonly used offset type — were likely "phantom credits" that did not represent genuine emissions reductions.

Who Was Exposed

Buyers of the affected credits for climate claims included Disney, Shell, Gucci, Salesforce, BHP, and easyJet.

The Specific Failure

The Kariba REDD+ project alone was later found by Verra itself to have issued roughly 15.2 million "excess" (over-issued) credits.

Verra's Response

Verra disputed the investigation's methodology, but announced it would phase out and replace its REDD+ methodology going forward.

The Market Impact

The controversy triggered a broader crisis of confidence in forest-based offsets, and accelerated demand for more rigorous, tech-verified credit types.

"You are a sustainability lead deciding whether to buy voluntary offsets after this scandal. What would you demand from a credit before trusting it?"

Connect this back to the four credibility tests from two slides ago.

🕒 3 MIN

SECTION 5 OF 7

India's Carbon Market — CCTS

India's own compliance market — built for a rapidly growing, developing economy.



From PAT to CCTS



How CCTS Actually Works

CCTS is deliberately built as two linked but distinct tracks.

The Compliance Mechanism

Energy-intensive sectors are given emissions-intensity targets (tCO₂e per unit of output) — not an absolute cap on total emissions.

Why Intensity, Not an Absolute Cap

India's economy is still growing fast. An intensity target lets output — and total emissions — rise while still forcing each unit of output to get cleaner every year.

The Offset Mechanism

Non-covered entities can register eligible projects (renewable energy, green hydrogen, landfill methane capture, and more) to earn tradable Carbon Credit Certificates (CCCs).

Over-Achievers vs. Under-Performers

Companies that beat their intensity target earn and can sell CCCs; companies that miss it must buy CCCs to cover the gap, or face a financial penalty.

Sectors & Institutions

Nine sectors, three institutions, one national registry.



Nine Covered Sectors

Aluminium, cement, chlor-alkali, fertiliser, iron & steel, pulp & paper, petrochemicals, petroleum refining, and textiles.



BEE (Administrator)

The Bureau of Energy Efficiency administers the scheme and sets technical standards and procedures.



Grid Controller of India

Operates the Indian Carbon Market registry, where every Carbon Credit Certificate is recorded and tracked.



CERC (Trading Oversight)

The Central Electricity Regulatory Commission oversees trading of CCCs on power exchanges (IEX, HPOWER, PXIL).

A Compliance Calculation Under CCTS

GIVEN

- A cement plant's FY target: 0.85 tCO₂e per tonne of cement produced
- Actual performance achieved: 0.80 tCO₂e per tonne of cement produced
- Annual cement production: 1,000,000 tonnes

FORMULA

$$\text{CCCs Earned} = (\text{Target Intensity} - \text{Actual Intensity}) \times \text{Production Volume}$$

STEP-BY-STEP CALCULATION

- 1 Intensity improvement: $0.85 - 0.80 = 0.05$ tCO₂e saved per tonne of cement
- 2 Total CCCs earned: $0.05 \times 1,000,000 = 50,000$ Carbon Credit Certificates
- 3 These 50,000 CCCs can now be sold on a power exchange to another entity that missed its own target.

50,000 CCCs

Carbon Credit Certificates earned by out-performing the FY intensity target — tradable for revenue on India's power exchanges.

Why does India's CCTS use emissions-intensity targets rather than an absolute emissions cap, unlike the EU ETS?

- A Intensity targets are easier for BEE to calculate
- B India has no interest in reducing emissions overall
- C It allows industrial output (and jobs) to keep growing while still forcing each unit of output to get cleaner ✓**
- D Absolute caps are illegal under Indian law

ANSWER & WHY

C. It allows industrial output (and jobs) to keep growing while still forcing each unit of output to get cleaner

India's economy is still industrializing and growing quickly. An absolute cap would constrain that growth; an intensity target lets total output rise while still requiring continuous efficiency improvement per unit produced — a design suited to a developing economy's priorities.

SECTION 6 OF 7

Risk Management in Carbon Markets

Every market this lecture has covered carries real financial and reputational risk — here is how to categorize it.

What Can Actually Go Wrong

Every carbon market participant — buyer, seller, or regulator — faces some mix of these four risk types.



Price / Market Risk

Allowance and credit prices can swing sharply on policy news, energy prices, or macroeconomic shocks.



Regulatory / Policy Risk

Rules change — caps tighten, methodologies get replaced, entire schemes get suspended or redesigned mid-cycle.



Delivery / Counterparty Risk

A seller may fail to deliver credits or allowances as promised, or a registry may cancel credits after purchase.

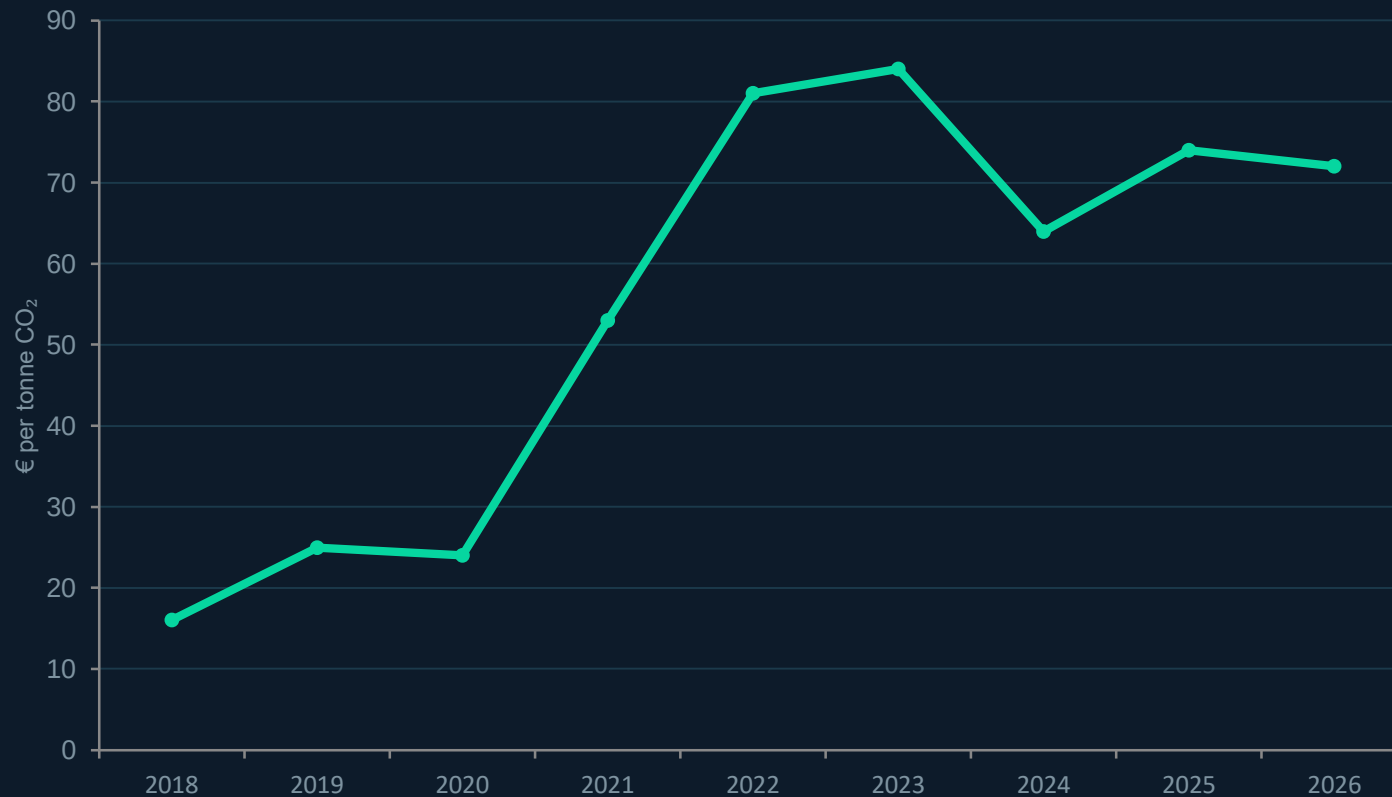


Reputational Risk

Buying low-quality or fraudulent credits and making public climate claims on them invites "greenwashing" accusations.

Reading a Real Carbon Price Chart

EU ETS allowance (EUA) price, illustrative annual trend — the same market referenced throughout this lecture.



Illustrative trend approximating published EUA price history (ICE/EEX, Sandbag Carbon Price Viewer).

WHAT THE SWINGS TELL YOU

- 2020: price dips as COVID-19 cuts industrial activity and emissions.
- 2021–2023: sharp rise as the cap tightens and the economy reopens.
- 2024: a notable pullback as emissions fall faster than expected.
- 2025–2026: recovery as front-loaded auctions end and CBAM tightens the market further.

Hedging Instruments in Carbon Markets

Large emitters rarely just buy allowances on the spot market — they manage exposure the way any commodity trader would.

Futures Contracts

Standardized exchange contracts (e.g. on ICE) to buy or sell allowances at a fixed price on a future date — locking in compliance cost in advance.

Forwards

Similar to futures but privately negotiated between two parties — more flexible, but carries more counterparty risk since it isn't exchange-cleared.

Options

The right (not the obligation) to buy or sell allowances at a set price — insurance against extreme price moves, paid for with an upfront premium.

Why This Matters

A cement plant with a €1M+ annual compliance cost cannot simply hope the price stays low — these instruments turn an unpredictable cost into a manageable, budgetable one.

Greenwashing Is Now a Legal and Financial Risk

The Verra case from Section 4 is a textbook example of reputational risk crystallizing into real financial and legal consequences.

1. The Claim

A company buys cheap rainforest credits and publicly claims "carbon neutral" status based on them.

2. The Exposure

Independent journalism or academic research finds the underlying credits don't represent real reductions.

3. The Consequence

Regulators, NGOs, and litigators increasingly treat unsupported climate claims as consumer-protection or securities violations.

4. The Lesson

Credit quality due diligence is no longer optional risk management — it is now core to avoiding legal and reputational exposure.

Which Risk Category Is This?

For each scenario, identify the primary type of carbon market risk it represents.

		PRICE RISK	REGULATORY RISK	REPUTATIONAL RISK
1	A government suddenly tightens the emissions cap mid-scheme			
2	EUA prices swing 20% in a month on energy market news	2	1	3
3	A company's "carbon neutral" claim is publicly debunked by journalists			
4	A new methodology replaces the one your credits were issued under			
5	Investors question a firm's climate credibility after a credit scandal			
6	A compliance cost budgeted at €70/tonne turns out to be €95/tonne	6	4	5

A trader buys the right, but not the obligation, to purchase EU allowances at €75/tonne before December — paying a premium today. What instrument is this?

A A futures contract

B A forward contract

C An option



D A spot trade

ANSWER & WHY

C. An option

An option grants the right — but not the obligation — to transact at a set price before a set date, in exchange for an upfront premium. Futures and forwards are firm commitments to transact; a spot trade settles immediately at today's price.

SECTION 7 OF 7

The Road Ahead

Where carbon markets are heading — and where your own skills fit into that future.



Article 6 & Linking Markets Together

The Paris Agreement includes a specific mechanism for countries to cooperate and trade emissions reductions across borders.

Article 6 of the Paris Agreement

Establishes rules for countries to transfer emissions reductions between each other, and for a new UN-supervised crediting mechanism.

Market Linking

Separate schemes (like the EU ETS and Switzerland's ETS) can be formally linked, letting allowances flow between them and converging their prices.

Why This Matters

As more national schemes like CCTS mature, the long-term direction is toward more interconnected, more liquid global carbon markets — with all the opportunity and risk that implies.

Carbon Markets Run on Infrastructure You Could Build

Every mechanism in this lecture — trading, verification, registries — is, underneath, a data and software problem.

MRV Technology

Satellite monitoring and remote sensing increasingly cross-check offset project claims — directly addressing the exact failure mode behind the Verra scandal.

Blockchain Registries

Distributed ledgers are being piloted to prevent double-counting and give buyers an auditable, tamper-evident credit history.

AI-Assisted Fraud Detection

Machine learning models can flag anomalous credit issuance patterns — the kind of over-issuance that took investigative journalists nine months to uncover by hand.

The Opportunity

Carbon market integrity is currently a trust problem waiting for better engineering — exactly the kind of gap this course keeps returning to.

"If you were building the fraud-detection system that could have caught the Verra over-issuance years earlier, what data would you want access to first?"

Think about what signals would have been visible before the journalists ever got involved.

🕒 3 MIN

The Complete Picture, Recapped

1

The Externality Problem

Carbon markets exist to make emitters pay the cost their pollution imposes on everyone else.

2

Tax vs. Cap-and-Trade

A tax fixes price and lets quantity float; cap-and-trade fixes quantity and lets price float.

3

The Social Cost of Carbon

A contested policy benchmark (~\$190/tonne per EPA), not a market price — and a lesson in how politicized this number is.

4

Cap-and-Trade Mechanics

Set the cap, allocate allowances, trade freely, then surrender allowances to prove compliance.

5

The EU ETS

The world's largest carbon market, now reinforced by CBAM at the EU's border.

6

Allowances vs. Credits

A legal permission to emit is not the same as a claimed reduction elsewhere — conflating them is the market's most common error.

7

The Verra Case

A reminder that "verified" is not automatically "real" — and that reputational risk is now a financial risk.

8

India's CCTS

An intensity-based compliance market plus a voluntary offset track, purpose-built for a growing economy.

9

Four Risk Categories

Price, regulatory, delivery/counterparty, and reputational risk — every participant faces some mix of all four.

10

Your Opportunity

MRV technology, blockchain registries, and AI fraud detection are all still being built — by people with exactly your skill set.